

Food Delivery

Req2Design · IEEE 830-1998 aligned

Document Control

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1: Introduction

1.1: Purpose

Purpose: To define the requirements for a food delivery application that allows customers to place orders, track their deliveries, and manage payment retries; restaurants to receive and prepare orders; drivers to deliver meals while handling route optimization and incident reporting; and the system to ensure reliable operation under various conditions.

1.2: Scope

Scope: This document covers the

specific requirements for the food delivery application including external interfaces, functionalities, non-functional attributes, and

system features without addressing security, scalability, maintainability, or availability aspects.

1.3: Definitions/Acronyms

- **API:** Application Programming Interface
- **DBMS:** Database Management System
- **UI:** User Interface

1.4: References

References: IEEE Std 830-1998

1.5: Overview

Overview: This SRS outlines the detailed specifications required for developing a robust food delivery platform encompassing customer ordering, restaurant management, driver operations, and system resilience.

2: Overall Description

2.1: Product Perspective

Product Perspective: The product is a web-based and mobile application designed for both customers and service providers (restaurants and drivers) to facilitate online food delivery services.

2.2: Product Functions

- **Customer functions:** Browsing restaurants, customizing orders, scheduling deliveries, tracking live rider locations, receiving status notifications.
- **Restaurant functions:** Receiving orders, confirming preparation times, marking orders as ready, pausing orders temporarily.
- **Driver functions:** Accepting delivery tasks, navigating optimized routes, reporting incidents, delivering meals.

External Interfaces:

- **User interface:** Web and mobile applications for customers, restaurants, and drivers.
- **Hardware interface:** Mobile devices and computers used by users.
- **Software interface:** Integration with third-party APIs for payment processing, mapping services, and notification systems.
- **Communication interface:** Real-time updates through push notifications and server-to-client communication.

2.4: Constraints

- Payment processing must automatically retry up to three times if initial attempts fail.
- Order substitution functionality must allow for easy replacement of unavailable items.

2.5: Assumptions/Dependencies

- Customers have access to internet-enabled devices.
- Restaurants provide accurate location information via GPS.
- Drivers possess valid licenses and are equipped with smartphones capable of running the app.

3: Specific Requirements

3.1: External Interface Requirements

3.1.1: User Interface

- Web and mobile applications accessible from browsers and native apps respectively.
- Intuitive design supporting multi-language options.
- Responsive layout ensuring compatibility across different screen sizes.

3.1.2: Hardware Interface

- Compatible with modern smartphones and tablets.

3.1.3: Software Interface

- Integrates with popular payment gateways.
- Supports real-time communication protocols for seamless interaction between client and server.

3.1.4: Communication Interface

- Push notifications for timely updates about order statuses.
- Server-to-client communication for real-time updates and interactions.

3.2: Functional Requirements

FR-01 Browse Restaurants

Input: None

Processing: Retrieve list of registered restaurants along with their menus.

Output: List of restaurants displayed with menu items.

Priority: High

FR-02 Customize Orders

Input: Selected restaurant ID and menu item IDs.

Processing: Display customizable options based on selected items.

Output: Customized order details shown to the user.

Priority: Medium

FR-03 Schedule Deliveries

Input: Time slot preferences provided by the user.

Processing: Validate available slots and suggest optimal delivery times.

Output: Confirmed delivery time communicated back to the user.

Priority: Low

FR-04 Track Live Rider Locations

Input: Active delivery task identifier.

Processing: Fetch current position data from the rider's device.

Output: Rider's location updated on the map.

Priority: High

FR-05 Receive Status Notifications

Input: Order ID or delivery task ID.

Processing: Monitor changes in order status and generate notifications.

Output: Timely alerts sent to relevant parties.

Priority: High

FR-06 Confirm Preparation Times

Input: Order ID.

Processing: Update estimated preparation duration upon receipt of confirmation from the restaurant.

Output: Updated expected delivery time reflected in the system.

Priority: Medium

FR-07 Mark Orders As Ready

Input: Order ID.

Processing: Acknowledge completion of meal preparation.

Output: Order marked as 'ready' in the system.

Priority: Medium

FR-08 Pause Orders Temporarily

Input: Order ID and reason for pausing.

Processing: Temporarily halt further processing until resumed.

Output: Order status updated accordingly.

Priority: Low

FR-09 Accept Delivery Tasks

Input: Available delivery task assigned by the system.

Processing: Assign the task to a driver and start navigation guidance.

Output: Route directions provided to the driver.

Priority: High

FR-10 Navigate Optimized Routes

Input: Current location of the driver and destination coordinates.

Processing: Calculate shortest path considering traffic and road conditions.

Output: Optimal driving directions displayed on the map.

Priority: High

FR-11 Report Incidents

Input: Incident type and description.

Processing: Log the event and notify appropriate authorities.

Output: Confirmation message indicating successful logging.

Priority: Medium

FR-12 Handle Order Substitutions

Input: Original order ID and new item(s) requested.

Processing: Replace original items with substitutes in the order record.

Output: Updated order details presented to the user.

Priority: Medium

3.3: Non-Functional Requirements

3.3.1: Usability

Usability: : Ensure all interfaces are intuitive and responsive, catering to diverse user needs.

3.3.2: Reliability

Reliability: : Implement mechanisms to prevent loss of critical data and ensure consistent performance.

3.3.3: Performance

Performance: : Minimize latency in response times to enhance user experience.

3.3.4: Portability

Portability: : Design the application to run efficiently on various platforms and devices.

4: System Features

Feature 1: Customer Ordering System

- Allows browsing of restaurants and viewing of menus.
- Enables customization of orders according to individual preferences.
- Provides scheduling capabilities for future deliveries.
- Tracks live rider positions and provides real-time updates.
- Sends timely notifications regarding order status.

Feature 2: Restaurant Management System

- Receives incoming orders directly from the application.
- Facilitates setting preparation times for orders.
- Offers a simple mechanism to pause orders when necessary.
- Displays order status updates promptly.

Feature 3: Driver Operations System

- Manages accepting delivery assignments from the system.
- Guides drivers through optimized routes using GPS technology.
- Supports reporting of any incidents encountered during delivery.
- Ensures smooth handover of completed deliveries.

Feature 4: Payment Retry Mechanism

- Automatically retries failed payment transactions up to three times.
- Handles order substitutions gracefully by allowing replacements within the same transaction.

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